

Martin Meinel

POSTDOCTORAL RESEARCHER · COMPUTATIONAL BIOLOGIST

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Profile

Computational biologist working with transcriptomic and genetic data for diagnostics and early drug target discovery. Reduced 20,000 genes to a two-gene diagnostic classifier, validated across 8 international cohorts and resulting in a filed patent and research-use-only diagnostic tool. Built an attention-based causal inference framework prioritizing candidate disease genes at single-cell-state resolution. Experienced in cross-functional collaborations spanning computational biology, clinical research, and experimental validation.

Research Experience

Gene-expression-based classifier enabling early rare cancer detection

Helmholtz Munich

PHD STUDENT

Oct 2021 - Aug 2026

- Developed a gene-expression-based diagnostic classifier to distinguish a rare cutaneous T cell lymphoma (early mycosis fungoides, MF) from the two most common inflammatory skin diseases (eczema and psoriasis).
- Created the biomarker discovery framework, SAFARI, revealing a **two-gene diagnostic signature** from bulk RNA-seq data comprising 20,000 protein-coding genes.
- Built a classifier with the option to defer clinically ambiguous cases. Achieved **91% sensitivity and 94% specificity** in the qPCR development cohort, and demonstrated generalizability across **8 independent international test cohorts**.
- Showed earlier detection than standard-of-care methods (histopathology and clinical presentation) in 4 longitudinally followed patients.
- Integrated public single-cell and spatial transcriptomic (10x Visium) datasets to identify tissue layers and cell types where biomarkers are expressed (T cells, keratinocytes).
- Filed **patent** covering the identified biomarkers and developed research-use-only diagnostic tool, Erysk, in collaboration with Dermagnostix.

FocusMR: Causal inference for early-stage drug target identification

Helmholtz Munich & CZI Residency

PHD STUDENT

Oct 2023 - Aug 2026

- Developed FocusMR, an attention-based Mendelian randomization framework that identifies candidate causal disease genes and the specific cell states in which they act, enabling more precise therapeutic targeting.
- Prioritized 34 candidate causal genes for eczema by integrating disease GWAS with population-scale single-cell genetics data (OneK1K), resolving **four loci to cell states finer than standard cell-type annotations**.
- Showed that the discovered cell states carry disease-associated biology by recovering known immune mechanisms, including PD-1 signaling, CD28 costimulation, and Th17/Th22 programs.
- Developed a Bayesian strategy to assess whether discovered cell states replicate in independent individuals.

Publications & Manuscripts

- Meinel, M.[†], Langreder, N.[†], et al. (2026). LCK and HOMER1 gene-expression based classifier distinguishes early mycosis fungoides from eczema and psoriasis. Manuscript completed peer review; in final editorial revision at *Science Translational Medicine*. preprint: <https://www.medrxiv.org/content/10.1101/2025.10.18.25338021v1>
- Garger, D.[†], Meinel, M.[†], Dietl, T., Hillig, C., Garzorz-Stark, N., Eyerich, K., de Angelis, M., Eyerich, S., Menden M.P. (2023). The impact of the cardiovascular component and somatic mutations on ageing. *Aging Cell*, <https://doi.org/10.1111/acer.13957>.
- Fischer, F., Doll, A., Uereyener, D., Roenneberg, S., Hillig, C., Weber, L., Hackert, V., Meinel, M., Farnoud, A., Seiringer, P., et al. (2023). Gene expression based molecular test as diagnostic aid for the differential diagnosis of psoriasis and eczema in formalin fixed and paraffin embedded tissue, microbiopsies and tape strips. *The Journal of Investigative Dermatology*, S0022-202X(23)00156-2. <https://doi.org/10.1016/j.jid.2023.02.015>.
- Farnoud, A., Ohnmacht, A. J., Meinel, M., & Menden, M. P. (2022). Can artificial intelligence accelerate preclinical drug discovery and precision medicine?. *Expert opinion on drug discovery*, 17(7), 661–665. <https://doi.org/10.1080/17460441.2022.2090540>

[†] symbolizes shared first authorship.

Skills

Genomic Technologies

Single-cell RNA-seq analysis & integration (scanpy, Harmony, scVI, scANVI, scCODA)
Spatial transcriptomics analysis & integration (scanpy, scANVI, 10x Visium)
Statistical genetics (Mendelian randomization, GWAS, HEIDI, Colocalization, PLINK)
Bulk RNA-sequencing analysis (DESeq2, edgeR, fgsea, clusterProfiler)

Programming & Data Analysis

Python (NumPy, Pandas, scikit-learn, PyTorch), R, Bash

High-Performance & Cloud Computing

Hydra, Slurm, Docker, Git, AWS

Languages

English (Fluent), German (Native)

Education

Postdoctoral researcher focusing on context-dependent eQTL mapping

HELMHOLTZ CENTER MUNICH | CASALE LAB

Munich, Germany

Aug 2026 - present

PhD in Computational Biology & Statistical Genetics

HELMHOLTZ CENTER MUNICH | LMU MUNICH | CHAN ZUCKERBERG INITIATIVE RESIDENCY

Munich, Germany

Oct 2021 - Aug 2026

- Supervised by Dr. Michael Menden, Dr. Natalie Garzorz-Stark & Dr. Francesco Paolo Casale

M.Sc. in Data Engineering & Analytics

TECHNICAL UNIVERSITY OF MUNICH

Munich, Germany

Oct 2018 - Nov 2020

B.Sc. in Applied Computer Science

UNIVERSITY OF BAYREUTH

Bayreuth, Germany

Oct 2014 - Oct 2017

Supervision, Teaching & Peer Review

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| 2025 | Supervised three students through their master's theses and internships. | Munich, Germany |
| 2025 | Reviewer for the RECOMB conference. | Seoul, South Korea |
| 2024 | Reviewer for the Machine Learning for Health (ML4H) workshop. | Vancouver, Canada |
| 2024 | Teaching assistant for "Computational Modeling for System Genetics" & "Machine Learning for Biologists". | TUM & LMU |

Honors & Awards

Patent

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| 2025 | Published patent application "Differential Diagnosis of Mycosis Fungoides: Method and System" (WO/2025/051923), using SAFARI to discover biomarkers distinguishing early MF from eczema & psoriasis. | Munich, Germany |
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Chan Zuckerberg Initiative Residency

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| 2024 | Selected for CZI residency for developing cell-context-aware drug target discovery methods to identify cellular contexts of causal genes for precise therapeutic targeting. | Redwood, USA |
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Oral presentations

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| 2025 | Oral presentation at NeurIPS AI4D3 workshop on how to identify cellular contexts of causal genes enhancing early-stage drug target discovery. | San Diego, USA |
| 2025 | Oral presentation at EORTC CLTG about the identification and clinical translation of biomarkers in early MF. | Athens, Greece |
| 2023 | Invited speaker session about "Biocomputational Analysis of Big Data" at the EADV-ESDR Summer Research Workshop: Big Data and Translation in Dermatology. | Freiburg, Germany |

Poster presentations

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| 2025 | Highlight poster with oral presentation on cell-state-aware single-cell Mendelian randomization at RECOMB Genetics. | Seoul, South Korea |
| 2025 | Presented work on cell-state-aware single-cell Mendelian randomization at the Helmholtz AI conference. | Karlsruhe, Germany |
| 2025 | Presented classifier development and validation results distinguishing early MF from psoriasis and eczema at DTKK meeting. | Freiburg, Germany |
| 2023 | Presented the SAFARI framework for rare disease biomarker discovery at the ML4H workshop. | New Orleans, USA |

Working experience

Opitz Consulting GmbH

SOFTWARE ENGINEER WORKING STUDENT

Munich, Germany

Oct 2019 - Oct 2020

- Built a Docker skeleton for Apache Zeppelin, enabling users to start a docker container running Apache Zeppelin on OpenShift with one click.
- Created a web application for Allianz IDS GmbH, allowing employees to track, assign, and assess investment orders within the department.

REHAU Industries SE & Co. KG

SOFTWARE ENGINEER WORKING STUDENT

Bayreuth, Germany

Nov 2017 - Mar 2019

- Developed an ALEXA skill for the 2018 FRONTALE window exhibition, demonstrating several features of the smart window.
- Implemented a web app for Medi Bayreuth to track player performance and fatigue for injury prevention.
- Developed an openHAB plugin to connect sensor-based garden watering systems with other IoT devices.